



Pipeline

Scrape HackerNews (Show/Top/New), Reddit (r/artificial, r/MachineLearning, r/singularity, r/LocalLLaMA), TechCrunch AI, VentureBeat AI, MIT Tech Review, The Verge AI, Wired AI, ArXiv CS.AI

Filter 40+ AI keyword patterns — LLMs, alignment, safety, research, tooling

Dedup Title-similarity (45% threshold) merges cross-source variants

Score Authenticity = 0.5 base + 0.3 per extra source + diversity + HN score bonuses

Rank Sorted by authenticity. Below 0.25 threshold are dropped.

Authenticity Tiers

- VERIFIED 0.80 - 1.00** 3+ independent sources including media
- CONFIRMED 0.50 - 0.79** 2 sources or 1 high-quality media
- EMERGING 0.25 - 0.49** Single-source; passes AI keyword filter

Source Breakdown

Source	Articles	Category
ArXiv CS.AI	209	Media
TechCrunch AI	12	Media
The Verge AI	10	Media
HackerNews	4	Community
MIT Tech Review	1	Media

VERIFIED INTELLIGENCE — 1 story

VERI

OpenAI launches Dots, its Muse competitor

The Verge AI +1 source

80%

OpenAI is responding to Meta's buzzy Muse AI with agentic helpers of its own: Dots. During its DevDay keynote on Tuesday, OpenAI announced that Dots will serve as always-on AI assistants that can "do nearly anything" across connected apps in the background while learning your...

<https://www.theverge.com/ai-artificial-intelligence/1002033/openai-dots-launch-muse-com...>

CONFIRMED SIGNALS — 234 stories

CONF

Show HN: TurboGPT: train 22KiB transformer in 13s

HackerNews 50% HN 49

Inspired by minGPT for home experiments. Requires CUDA 13.4; build script is Windows only.

<https://github.com/lostmsu/TurboGPT>

CONF

Show HN: Open-source skills for end-to-end video ad campaigns

HackerNews 50% HN 7

Hey there! We are building SuperCMO Skills, open source tools which allow agents (Claude Code, Cursor, Codex, etc.) to research, plan and create video and image campaigns end-to-end right out of your terminal or IDE. We got tired of paying for expensive AI subscriptions and...

<https://github.com/SupercmoHQ/superCMO-skills>

CONF

Show HN: Jauvex 1.2, two-way voice chat harness for Claude+Codex+Grok+Jev

HackerNews 50% HN 5

No summary — see link for full article.

<https://github.com/reindent/jauvex>

CONF

Show HN: Layout - Order food anywhere with AI

HackerNews 50% HN 2

Layout lets you order food anywhere with AI. Text it what you want, like "iced americano from Starbucks," and it builds the cart at the restaurant, sends you the total, and securely places the order. Works at almost any restaurant in the US over SMS, the iOS app, or any MCP...

<https://layout.link/>

CONF

The internet is convinced Elon Musk's xAI trolled OpenAI's 'Dots' launch

TechCrunch AI 50%

Before OpenAI launched its new AI agent, Dots, on Tuesday, Elon Musk's xAI had already acquired the domain name "dot.com," which now redirects to the Grok chatbot download page.

<https://techcrunch.com/2026/09/29/the-internet-is-convinced-elon-musks-xai-trolled-open...>

CONF

OpenAI's latest features take direct aim at the app store model

TechCrunch AI 50%

OpenAI is building out the pieces of an alternative to the traditional app store model, turning ChatGPT into a place where software can be discovered and used by people and AI agents alike.

<https://techcrunch.com/2026/09/29/openais-latest-features-take-direct-aim-at-the-app-st...>

CONF

Here's why OpenAI is absent from Nvidia's industry-wide effort to end rogue AI agents

TechCrunch AI 50%

OpenAI isn't a public supporter of Nvidia's Open Agent Safety Platform, but it is privately working with Nvidia, TechCrunch has learned.

<https://techcrunch.com/2026/09/29/heres-why-openai-is-absent-from-nvidias-industry-wide...>

CONF

OpenAI takes on Microsoft with the launch of what feels a whole lot like ChatGPT's own office suite

TechCrunch AI 50%

OpenAI's newly announced suite of office features puts it into more direct competition with more traditional software companies.

<https://techcrunch.com/2026/09/29/openai-takes-on-microsoft-with-the-launch-of-what-fee...>

0 CONF

AI-powered app maker Wabi pivots to a messaging experience

TechCrunch AI 50%

Wabi is repositioning its prompt-based app builder as a personal AI agent that can create interfaces on demand, combining chat, apps, and ongoing tasks.

<https://techcrunch.com/2026/09/29/ai-powered-app-maker-wabi-pivots-to-a-messaging-exper...>

1 CONF

OpenAI gives Codex reusable cloud environments that work across devices

TechCrunch AI 50%

OpenAI is expanding Codex with reusable cloud development environments, a revamped CLI with voice controls, new code review tools, and a security-focused product for scanning repositories and preparing fixes.

<https://techcrunch.com/2026/09/29/openai-gives-codex-reusable-cloud-environments-that-w...>

2 CONF

OpenAI expands ChatGPT's plug-ins with app-like interfaces and automations

TechCrunch AI 50%

OpenAI is expanding ChatGPT plug-ins with dedicated sidebar homes, interactive panels, file viewers, improved discovery, and support for automations.

<https://techcrunch.com/2026/09/29/openai-expands-chatgpts-plugins-with-app-like-interfa...>

3 CONF

Instinct founder said more than 50% of transactions on the platform are travel-related

TechCrunch AI 50%

Instinct founder said the platform is growing 10% day by day, with transaction volume increasing at a similar rate.

<https://techcrunch.com/2026/09/29/instinct-founder-said-more-than-50-of-transactions-on...>

4 CONF

Meta is expanding its AI agent Muse to small businesses

TechCrunch AI 50%

The tech giant says the agent can help owners run their business and find new customers.

<https://techcrunch.com/2026/09/29/meta-is-expanding-its-ai-agent-muse-to-small-businesses/>

5 CONF

OpenAI apologizes to Australia after its AI agents breached government sites

TechCrunch AI 50%

The company also detailed how some of those breaches had happened, and outlined additional measures it is taking to assess the impact of the events.

<https://techcrunch.com/2026/09/29/openai-apologizes-to-australia-after-its-ai-agents-br...>

6 CONF

Reco raises \$55M as AI agent security startups crowd the market

TechCrunch AI 50%

The round builds on a \$30 million fundraising in February, taking the company's total funding to \$140 million.

<https://techcrunch.com/2026/09/29/reco-raises-55m-as-ai-agent-security-startups-crowd-t...>

7 CONF

Making AI an asset, not an expense

MIT Tech Review 50%

When customers talk about AI costs, the conversation usually starts with token prices and ends with access to the latest, most capable model in the cloud. Do they always need that level of capability? Not necessarily. But that is often where the conversation goes. As AI moves...

<https://www.technologyreview.com/2026/09/29/1145186/making-ai-an-asset-not-an-expense/>

8 CONF

Sam Altman says OpenAI won't go public until its models are safe

The Verge AI 50%

For months, people have wondered when OpenAI will go public. CEO Sam Altman says it won't happen until the company can make better promises about model safety, with no firm timeline in sight. "We intend to continue with AI progress ... but as the models have had this surge forward..."

<https://www.theverge.com/ai-artificial-intelligence/1002505/sam-altman-openai-ipo-devda...>

9 CONF

Trump orders US government to call AI 'Super Intelligence'

The Verge AI 50%

The US executive branch is no longer acknowledging the existence of "artificial intelligence." Going forward, official policy websites, policy documents, and press releases will refer only to "Super Intelligence," thanks to a new executive order signed by President Donald Trump....

<https://www.theverge.com/policy/1002468/trump-ai-superintelligence-executive-order-ai>

0 CONF

Elon Musk's AI-powered Grokipedia is updating again

The Verge AI 50%

Grokipedia, the AI-powered online encyclopedia from SpaceXAI, appears to be updating articles once again after a months-long pause. In August, Lawfare reported that articles on Grokipedia hadn't reviewed edits since April, but the platform's live updates site is now showing...

<https://www.theverge.com/tech/1002448/elon-musk-grokipedia-ai-updating-again>

1 CONF

AI researchers put out videos saying superintelligence is 'exactly as dangerous as it sounds'

The Verge AI 50%

"The chance of human extinction is about a coin flip, in my view," Geoffrey Irving, a former OpenAI and Google DeepMind employee, said in a new interview. It's one of a dozen interviews with AI researchers, including current and former employees at OpenAI, Google, and Anthropic....

<https://www.theverge.com/ai-artificial-intelligence/1002238/openai-google-anthropic-ai-...>

2 CONF

Protesters gather at OpenAI's DevDay

The Verge AI 50%

On Tuesday, OpenAI's annual DevDay event began with protests, flyers, and chants. "Sam Altman, get off it, put people over profit," said a group of protesters marching in a circle in front of a series of signs that spelled out "PEOPLE OVER PROFIT." More than a dozen...

<https://www.theverge.com/ai-artificial-intelligence/1002201/openai-sam-altman-openai-de...>

3 **CONF****OpenAI DevDay 2026: The biggest news and announcements****The Verge AI** 50%

It's OpenAI's turn in the fall tech events calendar. The company is hosting its annual DevDay on September 29th in San Francisco, and during a live keynote featuring CEO Sam Altman, it revealed several notable updates, including Dots, its AI agent product that rivals Meta's...

<https://www.theverge.com/ai-artificial-intelligence/1001681/openai-devday-2026-biggest-...>
4 **CONF****Meta's Muse AI sent a YouTuber's address to a stranger****The Verge AI** 50%

Tech YouTuber Matt Robb says that Muse gave out his home address to a total stranger this weekend, after authorizing the bot to handle his Facebook Marketplace account. That's despite Meta placing great emphasis on the security features of Muse when it launched the personal AI...

<https://www.theverge.com/ai-artificial-intelligence/1001886/meta-muse-ai-facebook-marke...>
5 **CONF****Will Chinese AI companies slow down? A top House Democrat wants answers****The Verge AI** 50%

As President Donald Trump prepares to meet tech and AI CEOs in Washington, Rep. Ro Khanna (D-CA) is calling for a treaty between the US and China to keep AI from wreaking havoc on the world. But wrangling leaders in both countries to take action could be a long shot. In letters...

<https://www.theverge.com/policy/1001767/khanna-ai-safety-china-treaty>
6 **CONF****Anthropic warns of 'catastrophic' AI risks in its own IPO filing****The Verge AI** 50%

As Anthropic gears up for its greatly anticipated public debut, a preview of the company's IPO filing reportedly details its mounting losses, leadership proposals to retain power, and how its AI development plans could "further increase the risk that our models cause harm."...

<https://www.theverge.com/ai-artificial-intelligence/1001838/anthropic-ipo-prospectus-ai...>
7 **CONF****Towards Mitigating Deceptive Safety Alignment in Large Reasoning Models****ArXiv CS.AI** 50%

arXiv:2609.36254v1 Announce Type: new Abstract: Large Reasoning Models (LRMs) are commonly trained with reinforcement learning (RL) to improve their generation of chain-of-thought (CoT) reasoning before producing final answers. However, RL rewards are typically assigned based on...

<https://arxiv.org/abs/2609.36254>
8 **CONF****Neurosymbolic Routing for Reliable Reasoning on Resource-Constrained Edge Devices****ArXiv CS.AI** 50%

arXiv:2609.35833v1 Announce Type: new Abstract: Running a language model on edge hardware provides private and low-latency reasoning without a network connection, and yet the small models that fit on such devices are unreliable on the tasks computers are expected to handle well,...

<https://arxiv.org/abs/2609.35833>
9 **CONF****Fisher-Guided Progressive Parameter Selection for Adaptive Fine-Tuning****ArXiv CS.AI** 50%

arXiv:2606.10196v2 Announce Type: replace-cross Abstract: Parameter-efficient fine-tuning often selects trainable parameters before adaptation using architectural heuristics, without accounting for their varying importance during training. We introduce \textbf{FisherAdapTune},...

<https://arxiv.org/abs/2606.10196>

0 **CONF****More Programs or More Rolls? Separating Coverage from Specialization in LLM Harnesses****ArXiv CS.AI** **50%**

arXiv:2609.35873v1 Announce Type: new Abstract: Automated generation of LLM harnesses promises to improve inference through task specialization. Yet additional answer coverage can arise from repeated execution of the same program, making specialization difficult to identify. We...

<https://arxiv.org/abs/2609.35873>

1 **CONF****Beyond Symmetric Agents: Cognitive Diversity and Multi-Agent Debate in Small Language Models****ArXiv CS.AI** **50%**

arXiv:2609.35875v1 Announce Type: new Abstract: Multi-agent debate (MAD) reportedly improves reasoning and factuality over single-model inference, but prior work treats agents as symmetric peers, leaving open what drives the gains. We test the hypothesis that cognitive diversity...

<https://arxiv.org/abs/2609.35875>

2 **CONF****Representational Simplicity and Circuit Size Dissociate in a Threshold-Dependent Way: A Controlled Test via Adversarial Training****ArXiv CS.AI** **50%**

arXiv:2609.35890v1 Announce Type: new Abstract: Sparse-autoencoder decomposability and concentrated feature attribution are increasingly treated as evidence that a model's computation is easier to reverse-engineer. Whether this representational and attributional cleanliness...

<https://arxiv.org/abs/2609.35890>

3 **CONF****Less Uniform Discrete Diffusion is More Powerful and Scalable****ArXiv CS.AI** **50%**

arXiv:2609.35817v1 Announce Type: cross Abstract: Although uniform diffusion language models (UDLMs) represent a promising diffusion paradigm, scaling them remains challenging. We identify the core obstacle as an over-uniform training objective and condition-target confusion...

<https://arxiv.org/abs/2609.35817>

4 **CONF****Right Words, Wrong Moment: A Clinician-Grounded Analysis of Distress in 19,930 Conversations between Young People and ChatGPT****ArXiv CS.AI** **50%**

arXiv:2609.35953v1 Announce Type: new Abstract: Young people increasingly turn to General-Purpose Conversational Agents (GPCAs), such as ChatGPT, in moments of distress. We examine young adults' (ages 18-25) experiences using ChatGPT. We first collected 19,930 ChatGPT...

<https://arxiv.org/abs/2609.35953>

5 **CONF****State Trace Rationale As Auxiliary Task in Reinforcement Learning****ArXiv CS.AI** **50%**

arXiv:2609.36867v1 Announce Type: new Abstract: We propose STRAT, an auxiliary task that trains deep reinforcement learning (RL) agents to predict a short textual trace of their own state. Inspired by human spatial navigation, the description combines landmark, route, and survey...

<https://arxiv.org/abs/2609.36867>

- 6
CONF
Mirror-Score: Calibrated, Inference-only Scoring Exposes the Limits of Sequence-compatibility Ranking in D-peptide Design

ArXiv CS.AI 50%

arXiv:2609.36057v1 Announce Type: new Abstract: D-peptides combine protease resistance with high target specificity, but computational design of D-peptide binders remains immature. Mirror-Peptidizer introduced an in silico mirror-image screening pipeline using target reflection,...

<https://arxiv.org/abs/2609.36057>
- 7
CONF
A Polyphonic Conception of AI Understanding

ArXiv CS.AI 50%

arXiv:2609.36079v1 Announce Type: new Abstract: When a doctor, a judge, or an engineer must decide whether to trust an AI model's output, they cannot avoid asking what the model understands. Purely mathematical or statistical descriptions struggle to distinguish trustworthy from...

<https://arxiv.org/abs/2609.36079>
- 8
CONF
GeoOutageBench: Benchmarking Ambiguity-aware, Ontology-grounded Geospatiotemporal KGQA for Multimodal Power Outage and Resilience Analysis

ArXiv CS.AI 50%

arXiv:2609.36082v1 Announce Type: new Abstract: We introduce GeoOutageBench, a benchmark for assessing LLM-based geospatiotemporal KGQA for multimodal outage and resilience analysis. Unlike existing KGQA benchmarks for Web knowledge, GeoOutageBench considers a spatiotemporal KG...

<https://arxiv.org/abs/2609.36082>
- 9
CONF
An Exact Generate - Transform Decomposition of Small-LLM Team Scaling Across Orchestration Architectures

ArXiv CS.AI 50%

arXiv:2609.36104v1 Announce Type: new Abstract: Replacing one LLM agent with a collaborating team can raise accuracy, but whether scaling the team helps, and which architecture to scale, is unclear. Sweeping eight agent orchestration architectures across five instruction-tuned...

<https://arxiv.org/abs/2609.36104>
- 0
CONF
More Features Are Not More Evidence: Limits of Training-Free Human Activity Recognition with Jev

ArXiv CS.AI 50%

arXiv:2609.36154v1 Announce Type: new Abstract: General-purpose models promise sensor-based decisions without training a task-specific classifier, which could reduce the dependence of Human Activity Recognition (HAR) on labeled data. Yet it remains unclear whether such models...

<https://arxiv.org/abs/2609.36154>
- 1
CONF
Data-Free Pruning of Self-Attention Layers in LLMs

ArXiv CS.AI 50%

arXiv:2512.20636v2 Announce Type: replace-cross Abstract: Many self-attention sublayers in large language models (LLMs) can be removed with little to no loss. We attribute this to the Attention Suppression Hypothesis: during pre-training, some deep attention layers learn to mute...

<https://arxiv.org/abs/2512.20636>

2 **CONF****An Empirical Study and Assessment of EU AI Act Compliance Checkers****ArXiv CS.AI** **50%**

arXiv:2609.36228v1 Announce Type: new Abstract: The EU AI Act introduces extensive compliance requirements for organizations that develop, deploy, or integrate AI systems. Many of these requirements are directly relevant to security and privacy, while also addressing closely...

<https://arxiv.org/abs/2609.36228>

3 **CONF****Which Self-Improvements Should We Trust? Reliable Self-Improvement When Agents Reuse Their Benchmarks****ArXiv CS.AI** **50%**

arXiv:2609.33180v2 Announce Type: replace-cross Abstract: As recursive self-improvement (RSI) rapidly advances, reliable evaluation becomes critical for guiding adaptive search. RSI typically relies on finite evaluation resources, such as fixed benchmarks, to determine which...

<https://arxiv.org/abs/2609.33180>

4 **CONF****CoRe: Co-Evolving Reward Models for Mitigating Latent Reward Hacking in Video Diffusion Models****ArXiv CS.AI** **50%**

arXiv:2609.36245v1 Announce Type: new Abstract: Latent reward models (LRMs) enable efficient alignment of video diffusion models by scoring intermediate states directly in latent space. However, we find that optimizing against a fixed latent reward rapidly leads to latent reward...

<https://arxiv.org/abs/2609.36245>

5 **CONF****Train Ahead, Distill Back: Bootstrapping On-Policy Self-Distillation for Large Language Models****ArXiv CS.AI** **50%**

arXiv:2609.37132v1 Announce Type: new Abstract: On-policy self-distillation (OPSD) improves large language models by letting a self-teacher with privileged information provide dense token-level supervision on the model's own trajectories. Yet existing methods typically construct...

<https://arxiv.org/abs/2609.37132>

6 **CONF****Illusory Truth or Mere Exposure? Model-Dependent Repetition Effects in LLM-Based Social Media Simulations****ArXiv CS.AI** **50%**

arXiv:2609.36278v1 Announce Type: new Abstract: Generative agent-based models (GABMs) are increasingly used to simulate social media dynamics, including misinformation spread. For such social simulations to be valid proxies of human behavior, LLM agents should replicate...

<https://arxiv.org/abs/2609.36278>

7 **CONF****CheatBench: Measuring Reward Gaming in AI Agents****ArXiv CS.AI** **50%**

arXiv:2609.36308v1 Announce Type: new Abstract: Reinforcement learning has helped AI agents solve increasingly difficult tasks, but high rewards do not always reflect the work users intended. In recent incidents and controlled evaluations across the AI industry, agents trained...

<https://arxiv.org/abs/2609.36308>

8 CONF

HyperZip: Efficient Data Compression through Personalized Diffusion LLMs with Hypernetworks

ArXiv CS.AI 50%

arXiv:2609.36357v1 Announce Type: new Abstract: Large language models (LLMs) have shown strong potential for lossless data compression, but existing approaches are constrained by the high computational cost and low throughput of autoregressive decoding. We propose HyperZip, an...

<https://arxiv.org/abs/2609.36357>

9 CONF

Better Nearest Neighbor Graph Indices via (Efficient) LLM-Guided Pruning

ArXiv CS.AI 50%

arXiv:2609.36359v1 Announce Type: new Abstract: Graph-based approximate nearest neighbor search (ANNS) is widely used for large-scale semantic search. Its indices are constructed primarily based on geometric relationships among embeddings of an input dataset (e.g., documents or...

<https://arxiv.org/abs/2609.36359>

0 CONF

Engineering Simplicity: Simple Mechanism Interfaces Steer LLM Agents

ArXiv CS.AI 50%

arXiv:2609.36365v1 Announce Type: new Abstract: Can interaction formats and textual scaffolds help large language model (LLM) agents make better decisions, and do better decisions come with better explanations? We study these questions in auctions and matching, multi-agent...

<https://arxiv.org/abs/2609.36365>

1 CONF

Support-Set Target Leakage in Relational Foundation Models during In-Context Learning: Impact, Detection, and Mitigation

ArXiv CS.AI 50%

arXiv:2609.36384v1 Announce Type: new Abstract: Relational in-context learning (ICL) conditions predictions on the labeled support examples and their linked tables, creating a failure mode when the support set contains target-derived features that are unavailable for the query....

<https://arxiv.org/abs/2609.36384>

2 CONF

Persona Dosing: Calibrated Activation Steering for Graded Trait Control

ArXiv CS.AI 50%

arXiv:2609.36388v1 Announce Type: new Abstract: An activation-steering coefficient sets intervention strength, but requesting a particular degree of persona expression requires a behavioral scale. We study persona dosing: controlling a language model through a trait description...

<https://arxiv.org/abs/2609.36388>

3 CONF

BITEM at the NTCIR-19 R2C2 Task: Predicting Confidence from Agentic RAG Pipeline Signals

ArXiv CS.AI 50%

arXiv:2609.37993v1 Announce Type: cross Abstract: The BITEM team entered both subtasks of the NTCIR-19 R2C2 task with a single agentic pipeline, in which a model searches, reads and records evidence over a movie corpus while an orchestrator holds the record and rules on what may...

<https://arxiv.org/abs/2609.37993>

4 **CONF****Bits Under ZK-LLM: Evaluating Zero-Knowledge-Friendly Quantization for Verifiable Private LLM Inference****ArXiv CS.AI** **50%**

arXiv:2609.36437v1 Announce Type: new Abstract: Zero-knowledge proofs are emerging as a promising approach for enabling private, verifiable LLM governance and auditing, where regulators, users, and auditors need to verify claims about training-data usage or LLM inference-time...

<https://arxiv.org/abs/2609.36437>

5 **CONF****Implementing Cumulative Functions with Generalized Cumulative Constraints****ArXiv CS.AI** **50%**

arXiv:2508.01751v3 Announce Type: replace Abstract: Modeling scheduling problems with conditional time intervals and cumulative functions has become a common approach when using modern commercial constraint programming solvers. This paradigm enables the modeling of a wide range...

<https://arxiv.org/abs/2508.01751>

6 **CONF****Human-AI Collaboration: From Paradoxes to Patterns****ArXiv CS.AI** **50%**

arXiv:2609.36481v1 Announce Type: new Abstract: Evidence shows that humans and AI systems perform better together, by collaborating, than alone. This paper examines two key design dimensions of human-AI collaboration (autonomy and initiative) and explores the collaboration...

<https://arxiv.org/abs/2609.36481>

7 **CONF****Visual sensitivity is not claim retractability: persistence-aware credit assignment for multimodal reinforcement learning****ArXiv CS.AI** **50%**

arXiv:2609.36572v1 Announce Type: new Abstract: Reinforcement Learning with Verifiable Rewards (RLVR) has been extended to Large Vision-Language Models (LVLMs), and perception-aware methods further encourage policies to rely on visual evidence. Yet relying on the image does not...

<https://arxiv.org/abs/2609.36572>

8 **CONF****SafeCoEvo: Co-Evolving Safety Harnesses and Guards for LLM Agents at Test-Time****ArXiv CS.AI** **50%**

arXiv:2609.36580v1 Announce Type: new Abstract: LLM agents deployed in real-world environments continually encounter new tasks and safety risks, while execution feedback typically becomes available only after each task is completed. However, existing self-evolving approaches...

<https://arxiv.org/abs/2609.36580>

9 **CONF****Transformers Stop Thinking Too Early, and a Tiny LoRA Fixes It****ArXiv CS.AI** **50%**

arXiv:2609.36585v1 Announce Type: new Abstract: Pretrained transformers use little of their depth to follow references in context. Thirteen base models reliably follow only 1.4-3.6 lines, and extra pretrained loops add little. A task-trained rank-8 LoRA at one early layer...

<https://arxiv.org/abs/2609.36585>

- 0
CONF
Neural Structural Reasoner: A Brain-inspired Architecture for Reasoning over Structured Knowledge
ArXiv CS.AI 50%
arXiv:2609.36620v1 Announce Type: new Abstract: Structural reasoning, the ability to recognize and make inferences over the relational structure between objects and concepts, is a hallmark of human cognition, yet prevailing methods often collapse relational topology into flat...
<https://arxiv.org/abs/2609.36620>
- 1
CONF
DualTrack: Synchronized speech-gesture generation via symmetric coupling of pretrained priors
ArXiv CS.AI 50%
arXiv:2609.36624v1 Announce Type: new Abstract: Joint speech-gesture synthesis must coordinate two modalities despite limited paired data. Existing approaches often lack bidirectional interaction, have limited language coverage, or simplify body and finger representations. We...
<https://arxiv.org/abs/2609.36624>
- 2
CONF
ThinQuant: Scalable Rotation Learning for Weight and Activation Quantization of LLMs
ArXiv CS.AI 50%
arXiv:2609.36120v1 Announce Type: cross Abstract: Learned rotations play an important role in enabling low-bit weight and activation quantization of large language models by smoothing outliers in the activation distribution. State-of-the-art approaches include gradient-based...
<https://arxiv.org/abs/2609.36120>
- 3
CONF
MatToolBench: Benchmarking Multimodal Agents in Real-World Materials Science Workflows
ArXiv CS.AI 50%
arXiv:2609.37053v1 Announce Type: new Abstract: Multimodal GUI agents have achieved impressive results on general software benchmarks, yet their ability to operate professional scientific software remains largely unexplored. In materials science, sparse domain-specific web data,...
<https://arxiv.org/abs/2609.37053>
- 4
CONF
JudgeProfile: Understanding and Steering Subjectivity in LLM Judges
ArXiv CS.AI 50%
arXiv:2609.36705v1 Announce Type: new Abstract: LLM judges are inherently subjective, often favoring different responses in pairwise comparison when neither option is objectively wrong. To study this subjectivity, we introduce JudgeProfile, a framework that dissects LLM...
<https://arxiv.org/abs/2609.36705>
- 5
CONF
Can AI Scientists Change Their Minds? Prior-Evidence Conflict in Synthetic Universes
ArXiv CS.AI 50%
arXiv:2609.36726v1 Announce Type: new Abstract: Can a scientific agent distinguish a law it inferred from evidence from one it merely recognizes? We introduce Synthetic Universes, a controlled benchmark that pairs canonical famous worlds with matched twisted twins governed by...
<https://arxiv.org/abs/2609.36726>

6 CONF

BiFE: Search-Efficient Discovery of CPU-Only Branching Policies via LLM-based Bi-Fidelity Evolution

ArXiv CS.AI 50%

arXiv:2609.36735v1 Announce Type: new Abstract: In branch-and-bound (B&B) for mixed-integer linear programming (MILP), branching variable selection critically impacts efficiency. Existing neural branching policies often require GPU inference, while CPU-efficient symbolic...

<https://arxiv.org/abs/2609.36735>

7 CONF

AdviSD: Learning to Advise Frontier LLMs via Targeted Multi-Turn Self-Distillation

ArXiv CS.AI 50%

arXiv:2609.38142v1 Announce Type: new Abstract: A small trainable advisor can steer a frozen language-model executor using natural-language advice. In addition to learning from task rewards, the advisor can use feedback from completed interactions to improve its advice. However,...

<https://arxiv.org/abs/2609.38142>

8 CONF

AI as a Compiler: Compiling Triton kernels without the Triton compiler

ArXiv CS.AI 50%

arXiv:2609.36800v1 Announce Type: new Abstract: Compiler backends are expensive to build and maintain as programming models, workloads, and accelerators evolve. We investigate whether large language models can replace the conventional optimizing and lowering pipeline, a process...

<https://arxiv.org/abs/2609.36800>

9 CONF

IronLLM: Forging Compact Edge-Native Language Models for Real-Time Embodied Intelligence

ArXiv CS.AI 50%

arXiv:2609.36860v1 Announce Type: new Abstract: We present IronLLM-0.6B, a 654M-parameter language model designed for efficient on-device inference. IronLLM-0.6B combines a hybrid attention architecture with X-MTP, a lightweight shared-KV multi-token prediction design that...

<https://arxiv.org/abs/2609.36860>

0 CONF

TokenCast: Forecasting Token Consumption During LLM Agent Execution

ArXiv CS.AI 50%

arXiv:2609.35760v2 Announce Type: replace-cross Abstract: When a large language model (LLM) agent executes the same task, token consumption can vary by over an order of magnitude across runs. The agent chooses its next steps based on tool feedback and intermediate results, while...

<https://arxiv.org/abs/2609.35760>

1 CONF

Where Does Staleness Accumulate? Pool Aware Effective Staleness Control for Asynchronous RL in LLM Post-Training

ArXiv CS.AI 50%

arXiv:2609.36830v1 Announce Type: new Abstract: Fully asynchronous reinforcement learning (RL) improves resource utilization in large language model post-training by overlapping rollout generation with policy optimization, but it also introduces policy lag as trajectories are...

<https://arxiv.org/abs/2609.36830>

2 **CONF****Automated Screw Planning for Reduced Pelvic Fractures Based on Statistical Shape Models and Deep Learning**

ArXiv CS.AI 50%

arXiv:2609.36847v1 Announce Type: new Abstract: Percutaneous iliosacral screw fixation is an important minimally invasive treatment for unstable pelvic fractures. Because the sacroiliac region has complex anatomy and narrow screw corridors, the accuracy and safety of screw...

<https://arxiv.org/abs/2609.36847>

3 **CONF****When Upstream Messages Override Correct Answers: A Controlled Study of Multi-Agent LLM Collaboration**

ArXiv CS.AI 50%

arXiv:2609.36855v1 Announce Type: new Abstract: Multi-agent LLM systems rely on message passing among specialized agents to accomplish complex tasks. However, an upstream agent may provide useful information or an incorrect answer that causes a downstream agent to override a...

<https://arxiv.org/abs/2609.36855>

4 **CONF****Beyond Sub-Gaussian Detector Scores: Robust Weighted Profile-Loss Change Point Detection for Human-LLM Text Segmentation**

ArXiv CS.AI 50%

arXiv:2609.36888v1 Announce Type: new Abstract: Mixed human-LLM documents require locating authorship transitions from detector scores whose reliability varies across text units. Existing weighted mean contrasts are vulnerable to extreme scores, while directly replacing means...

<https://arxiv.org/abs/2609.36888>

5 **CONF****STAR-GRPO: Canonical Anchoring and Reliability-First Advantages against Representation-Dependent Reward Hacking**

ArXiv CS.AI 50%

arXiv:2609.36900v1 Announce Type: new Abstract: Reward hacking occurs when policy optimization exploits a brittle reward interface or an overly permissive proxy objective, improving the training score without improving the underlying response quality. This phenomenon is...

<https://arxiv.org/abs/2609.36900>

6 **CONF****From Checkpoint Variation to Selection Gains in Supervised Fine-Tuning**

ArXiv CS.AI 50%

arXiv:2609.36569v1 Announce Type: cross Abstract: Checkpoint selection is a routine decision in supervised fine-tuning (SFT): training produces multiple checkpoints, but only one is retained. Yet fixed-budget comparisons do not by themselves distinguish three empirical claims:...

<https://arxiv.org/abs/2609.36569>

7 **CONF****GradLev: Token-Parallel Test-Time Training Via Costate Prediction**

ArXiv CS.AI 50%

arXiv:2609.34174v2 Announce Type: replace-cross Abstract: Test-time training (TTT) allows a model to improve its predictions at inference time by updating weights after every observed token. However, sequential gradient writes make parallel training difficult. We observe that,...

<https://arxiv.org/abs/2609.34174>

8 CONF

Beyond Low-Rank Parameterization: Narrowing the Gap Between LoRA and Full Fine-Tuning via Gradient Decomposition

ArXiv CS.AI 50%

arXiv:2609.37027v1 Announce Type: new Abstract: Low-Rank Adaptation (LoRA) is a widely used approach to parameter-efficient fine-tuning (PEFT), yet a performance gap can remain relative to full fine-tuning (FFT). Many LoRA variants improve the initialization or optimization of...

<https://arxiv.org/abs/2609.37027>

9 CONF

Breaking the Illusion of Review Reliability under Static Evaluation: SCOPE Fuzzing for LLM-based Scientific Reviewers

ArXiv CS.AI 50%

arXiv:2609.37097v1 Announce Type: new Abstract: The rapid growth of submissions and reviewing workload has accelerated the use of large language models (LLMs) in peer review. Prior studies suggest that LLM-based reviewers can penalize content perturbations, such as overclaiming,...

<https://arxiv.org/abs/2609.37097>

0 CONF

Learning from Viable Failure Prefixes: Milestone Viability Potential Policy Optimization for Long-Horizon LLM Agents

ArXiv CS.AI 50%

arXiv:2609.37111v1 Announce Type: new Abstract: Long-horizon LLM agents require reinforcement learning methods that can assign credit to intermediate decisions under sparse and delayed rewards. Existing group-based methods such as GRPO and GiGPO alleviate this issue by comparing...

<https://arxiv.org/abs/2609.37111>

1 CONF

From Judgment Quality to Downstream Utility: Rethinking LLM-as-a-Judge for Open-Ended Tasks

ArXiv CS.AI 50%

arXiv:2609.37145v1 Announce Type: new Abstract: LLM-as-a-Judge is increasingly used to evaluate policy responses on open-ended tasks that lack ground-truth answers. Existing work often directly converts the resulting judgments into reward signals for policy training, paying...

<https://arxiv.org/abs/2609.37145>

2 CONF

SimpleEvol: An Agent-Loop Framework for LLM-Driven Automated Heuristic Design with Minimal Human Priors

ArXiv CS.AI 50%

arXiv:2609.37172v1 Announce Type: new Abstract: Large language models (LLMs) have emerged as powerful tools for automated heuristic design (AHD), enabling iterative generation and refinement of heuristics. However, the dominant paradigm embeds LLMs as narrow, fixed components,...

<https://arxiv.org/abs/2609.37172>

3 CONF

ARGOS: Reinforcement Learning-Driven Multidimensional Elasticity for Service Orchestration in the Computing Continuum

ArXiv CS.AI 50%

arXiv:2609.37085v1 Announce Type: cross Abstract: Data-intensive services in the Computing Continuum must balance analytics quality, resource usage, and cost across heterogeneous nodes with limited and uneven capacity. This balance becomes especially difficult when resource...

<https://arxiv.org/abs/2609.37085>

4 **CONF****OptiCom : A Unified Framework for State-Conditioned Composition in LLM-Driven Optimization****ArXiv CS.AI 50%**

arXiv:2609.37221v1 Announce Type: new Abstract: Large language models (LLMs) are increasingly deployed to solve complex scientific and practical problems via iterative optimization. However, dynamically coordinating diverse search mechanisms as candidate quality, failure modes,...

<https://arxiv.org/abs/2609.37221>

5 **CONF****MetaCtrl: Your Large Language Models Can Reason Better and More Concisely with a Metacognitive Controller****ArXiv CS.AI 50%**

arXiv:2609.37304v1 Announce Type: new Abstract: Large reasoning models improve performance on challenging problems by allocating additional computation before answering, but longer reasoning does not always lead to better results and can introduce substantial redundant reasoning...

<https://arxiv.org/abs/2609.37304>

6 **CONF****MMSkillRisk: Can Agents Stay Safe When Multimodal Skills Become Traps?****ArXiv CS.AI 50%**

arXiv:2609.35912v1 Announce Type: cross Abstract: Agent skills are shareable packages of procedural instructions, tools, and examples. Multimodal skills additionally include visual references that agents retrieve and inspect during execution. Because these images guide actions,...

<https://arxiv.org/abs/2609.35912>

7 **CONF****Teaching LLMs to Generate Challenging MILP Instances via Solver Feedback****ArXiv CS.AI 50%**

arXiv:2609.37356v1 Announce Type: new Abstract: Generating optimization instances that are both feasible and computationally challenging is crucial for benchmarking solvers and training learning-based optimization algorithms. Existing non-LLM generators rely on seed instances or...

<https://arxiv.org/abs/2609.37356>

8 **CONF****PHASE: A Physiology-Guided Hierarchical Foundation Model for Intracranial EEG****ArXiv CS.AI 50%**

arXiv:2609.36087v1 Announce Type: cross Abstract: Clinicians and neuroscientists have long analyzed intracranial electroencephalography (iEEG) through directly measurable physiological characteristics, which carry much of the information that downstream tasks depend on. Recent...

<https://arxiv.org/abs/2609.36087>

9 **CONF****Routing Should Pay for Itself: Sparse Supervision for Economical LLM Routing****ArXiv CS.AI 50%**

arXiv:2609.37402v1 Announce Type: new Abstract: Large language model (LLM) routing reduces serving cost by assigning each query to an appropriate model while preserving response quality. Learning such a router, however, often requires executing multiple candidate models on...

<https://arxiv.org/abs/2609.37402>

0 **CONF****VeriWeave Govern: Evidence-Gated Deterministic Runtime Governance for Enterprise AI Agents****ArXiv CS.AI 50%**

arXiv:2609.37457v1 Announce Type: new Abstract: Enterprise artificial-intelligence agents increasingly call tools, modify infrastructure, and process protected data, creating a need to separate action generation from action authorization. This article presents VeriWeave Govern,...

<https://arxiv.org/abs/2609.37457>

1 **CONF****CRASM-Gate: Deterministic-First Constraint- and Role-Aware Semantic Mapping with Selective Model Assistance Across Heterogeneous Industrial Standards****ArXiv CS.AI 50%**

arXiv:2609.37458v1 Announce Type: new Abstract: Industrial standards often encode the same engineering concept through incompatible hierarchies, identifiers, roles, and structural constraints, so the nearest lexical or embedding match can still be technically inadmissible. This...

<https://arxiv.org/abs/2609.37458>

2 **CONF****Authority Before Utility: Non-Compensatory Control for Persistent LLM Memory****ArXiv CS.AI 50%**

arXiv:2609.37474v1 Announce Type: new Abstract: Persistent memory creates a control problem that retrieval relevance alone does not solve: a memory can remain highly useful after an update, deletion, or revocation makes it inadmissible for the current answer. We formalize this...

<https://arxiv.org/abs/2609.37474>

3 **CONF****SkillGym: Training Skill-Use Agents with Automatic Verifiable Environment Generation****ArXiv CS.AI 50%**

arXiv:2609.37539v1 Announce Type: new Abstract: Skills equip LLM agents with professional knowledge and guidance to complete long-horizon and complex tasks. Although skills have been widely adopted in recent agent paradigms and harnesses, how to synthesize reliable training data...

<https://arxiv.org/abs/2609.37539>

4 **CONF****Flattening the Connectome Spectrum: A Spectral Filter for FC Induces a Pretraining Target for fMRI Encoders****ArXiv CS.AI 50%**

arXiv:2609.37642v1 Announce Type: new Abstract: Self-supervised pretraining reshaped prediction in language and vision, and brain foundation models (BFMs) inherited its promise. Representations learned from large unlabelled corpora should capture individual functional dynamics...

<https://arxiv.org/abs/2609.37642>

5 **CONF****MeanFlowAdvantage: Stable Reward Fine-Tuning for Few-Step Average-Velocity Generators****ArXiv CS.AI 50%**

arXiv:2609.37670v1 Announce Type: new Abstract: MeanFlow enables efficient few-step generation by predicting interval-average velocities, but this representation creates a mismatch for reward fine-tuning: existing advantage-based objectives are typically defined on instantaneous...

<https://arxiv.org/abs/2609.37670>

6 CONF

Locating Answer-Correctness Signals in Frozen Large Language Models

ArXiv CS.AI 50%

arXiv:2609.37700v1 Announce Type: new Abstract: Language models expose internal signals that predict whether an answer is correct, readable from a single forward pass of a frozen model without additional generations. Yet existing probes often commit to one signal family or layer...

<https://arxiv.org/abs/2609.37700>

7 CONF

Generative Interactions: Weaving Multiparty Human Motion with Bilevel Latent Dynamics

ArXiv CS.AI 50%

arXiv:2609.37708v1 Announce Type: new Abstract: Human social behaviour is not a collection of independent motions, but a jointly organised process in which group dynamics and individual variation continuously shape one another. Yet existing social motion models often prioritise...

<https://arxiv.org/abs/2609.37708>

8 CONF

Diffusion Reward Models

ArXiv CS.AI 50%

arXiv:2609.33803v2 Announce Type: replace-cross Abstract: Reward models underpin the alignment of large language models, yet the dominant designs reduce each prompt--response pair to a point estimate or to a distribution from a fixed parametric family. This is at odds with human...

<https://arxiv.org/abs/2609.33803>

9 CONF

A neural network that maintains and retrieves memories based on context

ArXiv CS.AI 50%

arXiv:2609.37791v1 Announce Type: new Abstract: Every day, people continuously infer situational context and adjust the way they understand and remember the world. Context, signaled by the prefrontal cortex, is known to modulate working memory and episodic memory, but the...

<https://arxiv.org/abs/2609.37791>

00 CONF

MoCA-Video: Motion-Aware Concept Alignment for Consistent Video Editing

ArXiv CS.AI 50%

arXiv:2506.01004v3 Announce Type: replace-cross Abstract: Unlike traditional video editing or inpainting, video semantic mixing fuses a reference concept with a moving target entity to produce a hybrid while preserving the source video's motion and layout. We propose MoCA-Video,...

<https://arxiv.org/abs/2506.01004>

01 CONF

Active Budget Can Kill Sensitivity: Diagnosing and Repairing TopK Sparse Autoencoder Reliability

ArXiv CS.AI 50%

arXiv:2609.37857v1 Announce Type: new Abstract: Sparse autoencoders (SAEs) are increasingly scaled to wider dictionaries to recover fine-grained structure from large language model activations. However, a feature is useful for interpretation only if it remains a stable unit of...

<https://arxiv.org/abs/2609.37857>

02 CONF

You Cannot Pick a Provider From the Price List: Market-Aware Routing for Open-Weight LLM Inference

ArXiv CS.AI 50%

arXiv:2609.37902v1 Announce Type: new Abstract: Existing LLM routers choose among models using static per-model costs. We show that open-weight inference markets introduce a second, largely ignored decision axis: after choosing a model, a client must still choose which provider...

<https://arxiv.org/abs/2609.37902>

03 CONF

GRFBrain: Graph-Structured Rectified Flows for EEG Dynamic Modeling

ArXiv CS.AI 50%

arXiv:2609.37934v1 Announce Type: new Abstract: Forecasting time-varying functional connectivity from electroencephalography (EEG) requires modeling both history-dependent trends and structured variability across channels. Conditional flow matching provides a framework for...

<https://arxiv.org/abs/2609.37934>

04 CONF

KV-Kaizen: Learning Context-Adaptive Cache Compression Choices

ArXiv CS.AI 50%

arXiv:2609.37988v1 Announce Type: new Abstract: As the context size of text processed with an LLM grows, the size of KV caches can outstrip the memory allocated for the original model weights. This impacts LLM throughput negatively, since decoding is memory-bound and decode cost...

<https://arxiv.org/abs/2609.37988>

05 CONF

HARISSA: Inference-Time Self-Checks for Efficient and Safe Local Language Model Deployment

ArXiv CS.AI 50%

arXiv:2609.38006v1 Announce Type: new Abstract: Running a language model locally offers advantages in privacy, latency, and cost, but local hardware fits only small models, which are less capable than frontier models. The usual remedy for a hard query, escalating it to a cloud...

<https://arxiv.org/abs/2609.38006>

06 CONF

Brain-SAD: A Brain-Inspired Safe Autonomous Driving Control Framework with Dynamic Fear-Oriented Constraint on Dual-Policy

ArXiv CS.AI 50%

arXiv:2609.38016v1 Announce Type: new Abstract: Constrained Reinforcement Learning has recently gained increasing attention in the field of Safe Autonomous Driving, where the general mechanism is to maximize the expected reward while keeping the overall action risk bounded. In...

<https://arxiv.org/abs/2609.38016>

07 CONF

PE-EK-PINN: Physics Embedding with Evolving Kernel for Scalable Physics-Informed Neural Networks

ArXiv CS.AI 50%

arXiv:2609.38023v1 Announce Type: new Abstract: Physics-Informed Neural Networks (PINNs) embed governing equations into deep learning, but enforce them only through loss residuals, leaving highly oscillatory wave behavior to be discovered by optimization. As a result, methods...

<https://arxiv.org/abs/2609.38023>

08 CONF

Probability is Not Enough: Exploring and Counting Divergent Tokens for Reasoning Uncertainty Quantification in LLMs

ArXiv CS.AI 50%

arXiv:2609.38070v1 Announce Type: new Abstract: As the chain-of-thought reasoning capabilities of large language models improve, evaluating and calibrating their reasoning confidence is becoming increasingly important for quantifying the uncertainty of their answers. Current...

<https://arxiv.org/abs/2609.38070>

09 CONF

Character Training for Risk-Averse Agents

ArXiv CS.AI 50%

arXiv:2609.38093v1 Announce Type: new Abstract: Risk aversion in resources could prevent misaligned AI agents from causing catastrophic harm. Misaligned but risk-averse agents would tend to favor safer strategies like making deals with humans over riskier strategies like...

<https://arxiv.org/abs/2609.38093>

10 CONF

Do LLM Agents Execute the Plans They Declare? From Planning-Mode Declaration to Pattern-Specific Execution

ArXiv CS.AI 50%

arXiv:2609.38108v1 Announce Type: new Abstract: Large language models (LLMs) enable agents to solve long-horizon tasks by generating a plan and then executing it in an environment. However, successful planning requires two distinct capabilities: selecting an appropriate plan for...

<https://arxiv.org/abs/2609.38108>

11 CONF

Thinking Before Thinking: Scaling Agentic Inference Through Meta-Reasoning

ArXiv CS.AI 50%

arXiv:2609.38147v1 Announce Type: new Abstract: As agents take on longer and more complex problems, controlling the execution becomes a task in its own right. Each step in the run brings new control choices, like which partial work to build on, whether to start fresh, or when to...

<https://arxiv.org/abs/2609.38147>

12 CONF

Bathtubs, Boundaries, and Sandboxes: AI Regulatory Learning under Legal Uncertainty

ArXiv CS.AI 50%

arXiv:2601.04094v4 Announce Type: cross Abstract: Effective regulation of AI is a defining policy challenge, driven by their integration into all aspects of society. To remain responsive to their rapid development and emergent properties, policymakers across the globe rely on...

<https://arxiv.org/abs/2601.04094>

13 CONF

Calibration-First Cross-Cohort Multimodal Temporal Learning for Transferable Asthma-Risk Forecasting

ArXiv CS.AI 50%

arXiv:2609.35795v1 Announce Type: cross Abstract: Asthma deterioration forecasting must remain reliable when patient populations, sensor ecosystems, and available modalities change across cohorts. Existing models commonly optimize within-cohort discrimination and may produce...

<https://arxiv.org/abs/2609.35795>

14 CONF

DraftTrace: A Multi-View Analytics Environment for AI-Integrated Writing

ArXiv CS.AI 50%

arXiv:2609.36544v1 Announce Type: cross Abstract: Generative AI has changed how students produce writing assignments. The final artifact is no longer sufficient to understand the process through which it was produced. We introduce DraftTrace, a writing environment that jointly...

<https://arxiv.org/abs/2609.36544>

15 CONF

Local Predictability and Collective Fidelity in LLM-Agent Societies

ArXiv CS.AI 50%

arXiv:2609.35813v1 Announce Type: cross Abstract: Compact surrogates could reduce the cost of simulating large language model societies, but must reproduce collective behavior. We compare individual predictions and collective forecasts using 9,455 published trajectories and new...

<https://arxiv.org/abs/2609.35813>

16 CONF

How to Run Statistics over LLM Judges and Trust the Results: Calibrated Inference for Small-Sample AI Evaluation with evalstats

ArXiv CS.AI 50%

arXiv:2609.35815v1 Announce Type: cross Abstract: Researchers across academia increasingly base significance claims on LLM judge scores and small-sample AI evaluations. Yet without well-calibrated confidence intervals (CIs), hypothesis tests, and judge-bias corrections, such...

<https://arxiv.org/abs/2609.35815>

17 CONF

When Should LLMs Trust Their Own Revisions? A Risk-Aware Study of Intrinsic Self-Correction

ArXiv CS.AI 50%

arXiv:2609.35832v1 Announce Type: cross Abstract: Intrinsic self-correction asks a language model to revise its own answer without receiving new external evidence. A second pass can recover mistakes, but it can also overturn answers that were already correct. We study this...

<https://arxiv.org/abs/2609.35832>

18 CONF

Mara Chain: Rethinking Failure as a Stepping Stone for AI System Auto-Evolution

ArXiv CS.AI 50%

arXiv:2609.35855v1 Announce Type: cross Abstract: Optimizing deployed AI systems increasingly amounts to editing prompts, skills, harnesses, and code rather than model weights. Existing approaches commonly optimize these artifacts through propose-evaluate-select procedures,...

<https://arxiv.org/abs/2609.35855>

19 CONF

Beyond Keywords: Leveraging Generative LLMs and Label Aggregation to Classify Economic Policy Uncertainty in News Articles

ArXiv CS.AI 50%

arXiv:2609.35856v1 Announce Type: cross Abstract: This research describes the adaptation of Large Language Models (LLMs) for economic monitoring in the public sector to automatically determine whether an article discusses Economic Policy Uncertainty (EPU) and to identify its...

<https://arxiv.org/abs/2609.35856>

20 **CONF****SINGED: Correct Outputs Do Not Certify Safe Execution in LLM Agents**

ArXiv CS.AI 50%

arXiv:2609.35889v1 Announce Type: cross Abstract: Tool-using language-model agents select and execute third-party artifacts. Different implementations can return the requested output while producing hidden execution effects that task-, attack-, or choice-based evaluations may...

<https://arxiv.org/abs/2609.35889>
21 **CONF****Reconstructing Implicit Scientific Knowledge: Evaluating LLM Agents through End-to-End Reproduction of Astronomy**

ArXiv CS.AI 50%

arXiv:2609.35900v1 Announce Type: cross Abstract: The integration of large language models (LLMs) into scientific workflows is accelerating, yet their ability to reconstruct the reasoning underlying published research remains unexplored. Papers specify explicit procedures while...

<https://arxiv.org/abs/2609.35900>
22 **CONF****Similarity Is Not Validity: Defending LLM Semantic Caches Against Poisoning**

ArXiv CS.AI 50%

arXiv:2609.35908v1 Announce Type: cross Abstract: Semantic caches reduce LLM serving costs by reusing previously generated answers for semantically similar queries. However, retrieval is based solely on embedding similarity between the incoming query and cached queries. This...

<https://arxiv.org/abs/2609.35908>
23 **CONF****Evaluating Test-Time Scaling of General LLM Agents**

ArXiv CS.AI 50%

arXiv:2602.18998v2 Announce Type: replace Abstract: LLM agents are increasingly expected to operate as general-purpose systems that resolve real-world user requests, yet their dynamic scaling behavior in realistic environments remains poorly understood. In this paper, we...

<https://arxiv.org/abs/2602.18998>
24 **CONF****Graph neural networks for sampling-invariant embeddings of organized signal sets**

ArXiv CS.AI 50%

arXiv:2609.35934v1 Announce Type: cross Abstract: Sensor networks and radars can deliver signals as organized sets, e.g. ordered signals, signals describing range cells within a grid or signals perceived as graph nodes. Within such sets, individual signals may be characterized...

<https://arxiv.org/abs/2609.35934>
25 **CONF****Embodied Semantic Communication for Collective Autonomous Agents: A Tutorial on Representation, Wireless Delivery, and Closed-Loop Coordination**

ArXiv CS.AI 50%

arXiv:2609.35936v1 Announce Type: cross Abstract: As autonomous systems and embodied intelligence enter the dynamic physical world, multi-agent collaboration calls for a paradigm shift in communication design. However, existing communication paradigms overlook that agents form...

<https://arxiv.org/abs/2609.35936>

26 CONF

HEAR: Real Voices, Real Bias: A Large-Scale Human-Recorded, Demographically Diverse Benchmark for Audio Language Models

ArXiv CS.AI 50%

arXiv:2609.35952v1 Announce Type: cross Abstract: We introduce HEAR (Human-recorded Evaluation of Audio-LLM bias by Real speakers), a large-scale, ecologically valid benchmark comprising 87k real human audio samples from 843 demographically diverse participants. HEAR enables...

<https://arxiv.org/abs/2609.35952>

27 CONF

Solver Agent: an Agentic AI Framework for Theoretical Physics Computations Applied to F-theory Uplifts of O3-planes and S-folds

ArXiv CS.AI 50%

arXiv:2609.35958v1 Announce Type: cross Abstract: We introduce Solver Agent, an AI framework based on large language models for calculations and proofs in mathematics and theoretical physics. The solution process is tracked through a persistent ledger that records assumptions,...

<https://arxiv.org/abs/2609.35958>

28 CONF

Systematic Multi-Agent Vision-and-Language Navigation: Formulation, Benchmark, and Method

ArXiv CS.AI 50%

arXiv:2609.35965v1 Announce Type: cross Abstract: Vision-and-Language Navigation (VLN) has largely focused on a single agent following a single instruction, yet many real-world applications require teams of robots to tackle tasks beyond the capabilities of any individual agent....

<https://arxiv.org/abs/2609.35965>

29 CONF

Causal and Interpretable Structures in LLM Compositional Tasks

ArXiv CS.AI 50%

arXiv:2609.35970v1 Announce Type: cross Abstract: Large language models are able to solve tasks whose answers depend on not only individual input tokens, but also on relations among them. How is such relational information represented and processed across transformer layers? We...

<https://arxiv.org/abs/2609.35970>

30 CONF

Infrared Subtraction with Artificial Intelligence

ArXiv CS.AI 50%

arXiv:2609.36007v1 Announce Type: cross Abstract: We present AI-developed local infrared subtraction, building on projection to Born and EFT matching. The framework separates an integrable radiation term from a finite contribution at Born kinematics, referred to as the Born...

<https://arxiv.org/abs/2609.36007>

31 CONF

Improving scalable oversight with co-trained monitors

ArXiv CS.AI 50%

arXiv:2609.36049v1 Announce Type: cross Abstract: Worker-monitor setups are a promising approach to AI oversight, but training workers against fixed monitors can incentivize monitor evasion. We study whether this failure mode can be avoided by co-training the monitor alongside...

<https://arxiv.org/abs/2609.36049>

32 CONF

What if automating AI R&D triggers an intelligence explosion?

ArXiv CS.AI 50%

arXiv:2609.36054v1 Announce Type: cross Abstract: In contrast to even a year ago, AI systems now write most of the code inside the companies that build them. As more of the AI research and development (R&D) pipeline is automated, could AI progress radically accelerate in an...

<https://arxiv.org/abs/2609.36054>

33 CONF

FLOOR: A Human-Aligned Domain-Specific Language Model for Architectural Design

ArXiv CS.AI 50%

arXiv:2609.36064v1 Announce Type: cross Abstract: Foundation models are powerful generators, but many engineering domains require structured representations that general-purpose systems handle poorly. We introduce FLOOR (Floor Layout Optimization with RL Alignment), a family of...

<https://arxiv.org/abs/2609.36064>

34 CONF

AerialDojo-200K: A Large-Scale Benchmark Suite for Open-World Aerial Object-Goal Search

ArXiv CS.AI 50%

arXiv:2609.36066v1 Announce Type: cross Abstract: Open-world aerial object-goal search is a foundational yet challenging task, requiring aerial agents to autonomously explore large-scale, unstructured three-dimensional environments and reach target objects specified by semantic...

<https://arxiv.org/abs/2609.36066>

35 CONF

The Uneven Decline of Collective Knowledge Production: Evidence from Stack Overflow After Generative AI

ArXiv CS.AI 50%

arXiv:2609.36069v1 Announce Type: cross Abstract: Generative AI (Gen AI) is reshaping how individuals learn and work, but its consequences for collective knowledge, the shared body of knowledge that online communities produce together, remain poorly understood. Prior work has...

<https://arxiv.org/abs/2609.36069>

36 CONF

Measuring trainable degrees of freedom in materials graph neural networks: a random-subspace intrinsic dimension analysis

ArXiv CS.AI 50%

arXiv:2609.36084v1 Announce Type: cross Abstract: Final predictive accuracy is the standard basis for comparing graph neural networks (GNNs) in materials-property prediction, but it does not show how strongly performance depends on access to trainable parameter-space directions...

<https://arxiv.org/abs/2609.36084>

37 CONF

PADM\': Preference Alignment Data Synthesis for Meta-Evaluation of LM Agent Evaluators

ArXiv CS.AI 50%

arXiv:2609.36086v1 Announce Type: cross Abstract: Language models are frequently employed to evaluate other language models. An LM evaluator scoring agentic behaviors across multiple criteria is valuable, provided that its decisions align with human judgment. We call the problem...

<https://arxiv.org/abs/2609.36086>

38 CONF

Similar Choices, Different Attention: Cross-Modal Associations in Humans and Vision-Language Models

ArXiv CS.AI 50%

arXiv:2609.36475v1 Announce Type: cross Abstract: Cross-modal associations are systematic pairings of features across modalities, such as the association of 'bouba' with round shapes and 'kiki' with sharp shapes. Prior work has compared humans and vision-language models (VLMs)...

<https://arxiv.org/abs/2609.36475>

39 CONF

LoopICL: Looping a single transformer block to solve tabular tasks

ArXiv CS.AI 50%

arXiv:2609.36108v1 Announce Type: cross Abstract: Tabular foundation models using in-context learning have recently surpassed gradient-boosted trees on predictive tabular tasks. However, recent mechanistic insights suggest that parameters in these models are largely redundant....

<https://arxiv.org/abs/2609.36108>

40 CONF

Accessible, but Not Adopted: Increasing LLM Adoption among First-generation, Low-income (FGLI) College Students beyond Expanding Access

ArXiv CS.AI 50%

arXiv:2609.36129v1 Announce Type: cross Abstract: Large language models (LLMs) are increasingly positioned as a force to empower underserved communities, and significant efforts are being made to expand access. Yet, access alone does not equate to meaningful adoption. First,...

<https://arxiv.org/abs/2609.36129>

41 CONF

CoMemBench: Benchmarking Collaborative Memory Boundaries across Multi-Agent Workflow Topologies

ArXiv CS.AI 50%

arXiv:2609.32192v2 Announce Type: replace Abstract: Multi-agent workflows require task-relevant information to be shared across agents, while irrelevant, stale, unverified, or incompatible information must remain isolated. We call this task-conditioned scope of information a...

<https://arxiv.org/abs/2609.32192>

42 CONF

What Does Post-Training Change in Multilingual Reasoning?

ArXiv CS.AI 50%

arXiv:2609.37104v1 Announce Type: cross Abstract: Open-source reasoning models provide unequal access to reasoning capability across languages. When a model can solve a problem but cannot deliver a complete solution in the user's language, language becomes an access barrier...

<https://arxiv.org/abs/2609.37104>

43 CONF

Population Fidelity: Evaluating Population Representativeness in LLMs

ArXiv CS.AI 50%

arXiv:2609.36253v1 Announce Type: cross Abstract: Large language models (LLMs) show considerable potential in simulating human attitudes and preferences. Prior work finds that LLM-generated responses can compress the range of attitudes found within populations and misrepresent...

<https://arxiv.org/abs/2609.36253>

44 CONF

Paired Multimodal Scaling Laws

ArXiv CS.AI 50%

arXiv:2609.36263v1 Announce Type: cross Abstract: Existing multimodal scaling laws fit multimodality terms empirically after testing and never vary how much data is multimodally paired at fixed data budgets. We investigate how, under the same total data per modality, changing...

<https://arxiv.org/abs/2609.36263>

45 CONF

Proofs Without Nominals: Gödel's Ontological Argument, its Shallow Embedding, and the Open Questions of the Monatshefte Notes

ArXiv CS.AI 50%

arXiv:2609.36279v1 Announce Type: cross Abstract: The shallow embedding of higher-order modal logic in classical higher-order logic, used in Benzmüller and Scott's Notes on Gödel's and Scott's variants of the ontological argument (2025), reaches beyond the modal object...

<https://arxiv.org/abs/2609.36279>

46 CONF

How Much Prompt Is Enough? A Blackbox Minimization of Few-Shots in LLMs

ArXiv CS.AI 50%

arXiv:2609.36289v1 Announce Type: cross Abstract: Prompts are the primary mechanism for directing the behavior of large language models (LLMs). Yet the internal structure and causal hierarchy of prompts remain poorly understood: which parts are causally necessary and which are...

<https://arxiv.org/abs/2609.36289>

47 CONF

Beyond Semantics: How Temporal Biases Shape Retrieval in Transformer and State-Space Models

ArXiv CS.AI 50%

arXiv:2510.22752v2 Announce Type: replace-cross Abstract: In-context learning is governed by both temporal and semantic relationships, shaping how Large Language Models (LLMs) retrieve contextual information. Analogous to human episodic memory, where the retrieval of specific...

<https://arxiv.org/abs/2510.22752>

48 CONF

Calibrating One-Round Membership Inference with Neighbors

ArXiv CS.AI 50%

arXiv:2609.36331v1 Announce Type: cross Abstract: The state-of-the-art Membership Inference (MI) methods calibrate their signal separately for each example using reference models, auxiliary models trained to exclude the target. This paradigm scales poorly to modern large models,...

<https://arxiv.org/abs/2609.36331>

49 CONF

Representation by Design in Generation: Cross-View Class-Token Alignment in Diffusion Transformers

ArXiv CS.AI 50%

arXiv:2609.36348v1 Announce Type: cross Abstract: Generative and representation learning remain asymmetrically connected: semantic representations are used to improve diffusion generation, whereas the models' own representations are often treated as a by-product of synthesis. We...

<https://arxiv.org/abs/2609.36348>

50 **CONF****Explainability from Training with Applications to TCR-Epitope Prediction****ArXiv CS.AI 50%**

arXiv:2609.36354v1 Announce Type: cross Abstract: Deep learning models have achieved strong performance in artificial intelligence for science, yet their black-box nature limits our understanding of how they learn scientific tasks. Existing methods for interpretability provide...

<https://arxiv.org/abs/2609.36354>

51 **CONF****ThinkingGuard: Decoding Implicit Hazards via Step-by-Step Risk Attribution in Multimodal Large Language Models****ArXiv CS.AI 50%**

arXiv:2609.36562v1 Announce Type: cross Abstract: While Multimodal Large Language Models (MLLMs) are increasingly deployed in safety-critical domains, their reliability is threatened by multimodal implicit risks. Unlike explicit threats, these hazards emerge when individually...

<https://arxiv.org/abs/2609.36562>

52 **CONF****Quantization Enables Private Dense Retrieval against Malicious Service Providers****ArXiv CS.AI 50%**

arXiv:2609.36376v1 Announce Type: cross Abstract: Dense retrieval, the key component of Retrieval Augmented Generation (RAG), retrieves the most relevant documents by comparing dense vector representations of queries and passages from a large corpus. In privacy-sensitive...

<https://arxiv.org/abs/2609.36376>

53 **CONF****FineART: Fine-grained Annotated Robotic Trajectory Dataset and Vision-Language-Action Model for Bimanual Manipulation****ArXiv CS.AI 50%**

arXiv:2609.36416v1 Announce Type: cross Abstract: Robots operating in real-world environments must execute complex, multi-step bimanual tasks over long horizons rather than single, isolated actions. Current manipulation datasets struggle to support this capability: although...

<https://arxiv.org/abs/2609.36416>

54 **CONF****Online Versatile Incremental Learning: Towards Class and Domain-Agnostic Adaptation at Any Time****ArXiv CS.AI 50%**

arXiv:2609.36442v1 Announce Type: cross Abstract: Continual learning enables vision systems to adapt to ever-changing data distributions. Despite significant advances, existing approaches fail to capture continuous and concurrent shifts in classes and domains, a critical...

<https://arxiv.org/abs/2609.36442>

55 **CONF****Guard Models Are Overconfident Where Base Models Are Uncertain****ArXiv CS.AI 50%**

arXiv:2609.36477v1 Announce Type: cross Abstract: Guard models are used as safety classifiers, with confidence scores driving downstream moderation decisions. We evaluate five guard models for prompt classification and find that although several are nearly calibrated on clean...

<https://arxiv.org/abs/2609.36477>

56 CONF

Towards Breaking the Learning System Wall Using Multimodal Tutoring Transcriptions

ArXiv CS.AI 50%

arXiv:2609.36502v1 Announce Type: cross Abstract: Past research using log data has faced the "learning system wall," whereby few methods exist for generalizing models of student learning across platforms. Increasingly, online learning is captured by richer forms of data,...

<https://arxiv.org/abs/2609.36502>

57 CONF

WitnessGym: Benchmarking Coding Agents on the Construction of Bug Witnesses

ArXiv CS.AI 50%

arXiv:2609.36635v1 Announce Type: cross Abstract: Bug validation asks a coding agent to produce an executable witness for a reported bug. The witness combines a concrete input with a testing harness and exposes faulty behavior during execution. Such evidence makes audit findings...

<https://arxiv.org/abs/2609.36635>

58 CONF

AutoLoCo: Communication Efficient Distributed LLM Training via Adaptive Synchronization

ArXiv CS.AI 50%

arXiv:2609.36662v1 Announce Type: cross Abstract: The pre-training of Large Language Models (LLMs) is increasingly conducted across multiple data centers. As training scales to a larger number of accelerators, the fraction of time spent on computation decreases, while the...

<https://arxiv.org/abs/2609.36662>

59 CONF

Normalize-Then-Precondition: A Hierarchical Approach to Marginal Scale and Interaction Geometry for LLM Training

ArXiv CS.AI 50%

arXiv:2609.36692v1 Announce Type: cross Abstract: Matrix optimizers have emerged as a promising direction, with Muon standing out as a prominent design. Revisiting Muon through its full-Gram representation, we observe that it jointly processes marginal-scale and interaction...

<https://arxiv.org/abs/2609.36692>

60 CONF

Does the Unsafe Gradient Survive a Conversation? On the Fragility of Gradient-Based Jailbreak Detection in Multi-Turn Dialogue

ArXiv CS.AI 50%

arXiv:2609.36849v1 Announce Type: cross Abstract: Safety-aligned language models are commonly deployed as multi-turn assistants, which lets adversaries spread unsafe intent across several user turns instead of a single prompt. Gradient-based jailbreak detectors such as GradSafe...

<https://arxiv.org/abs/2609.36849>

61 CONF

Frontier Autolab: Organizational Memory, Adversarial Dissent and Temporal Leakage in Multi-Agent LLM Firms Across Fifty Years of Technological Change

ArXiv CS.AI 50%

arXiv:2609.36739v1 Announce Type: cross Abstract: Multi-agent LLM systems are increasingly structured like organizations, with roles, critics and shared memory, yet they are evaluated on tasks that last minutes. We ask how such an organization behaves when the ground it stands...

<https://arxiv.org/abs/2609.36739>

62 CONF

Diffusion Policy Improvement with Proposal-Conditioned Refinement Flows

ArXiv CS.AI 50%

arXiv:2609.36812v1 Announce Type: cross Abstract: Diffusion and flow policies can model complex behaviors in offline reinforcement learning (RL). However, penalizing their KL divergence from the behavior policy can discourage actions having high critic values with low behavior...

<https://arxiv.org/abs/2609.36812>

63 CONF

Safer Content or Firmer Refusals? A Hybrid Perturbation Defense for Alignment under Harmful Fine-tuning

ArXiv CS.AI 50%

arXiv:2609.36862v1 Announce Type: cross Abstract: Fine-tuning-as-a-service lets users adapt a safety-aligned language model to their own data, but it also creates a harmful fine-tuning attack surface: a small amount of harmful data mixed into an otherwise benign fine-tuning set...

<https://arxiv.org/abs/2609.36862>

64 CONF

SKILLITE: Evidence-Guided Malicious Skill Auditing with Compact LLMs

ArXiv CS.AI 50%

arXiv:2609.36879v1 Announce Type: cross Abstract: As LLM-based agents perform increasingly complex tasks, Agent Skills have emerged as a flexible mechanism for extending their capabilities. An Agent Skill packages task-specific instructions with executable components and...

<https://arxiv.org/abs/2609.36879>

65 CONF

State Transport Routing for Short-horizon Adaptation in Multi-horizon Photovoltaic Forecasting

ArXiv CS.AI 50%

arXiv:2609.36926v1 Announce Type: cross Abstract: Recent power measurements provide valuable information for photovoltaic(PV) power forecasting, but directly extrapolating short-term trends can introduce substantial errors over longer forecast horizons. To address this...

<https://arxiv.org/abs/2609.36926>

66 CONF

ER-JEPA: Experience Replay Improves Joint-Embedding Predictive Learning in Language Models

ArXiv CS.AI 50%

arXiv:2609.36952v1 Announce Type: cross Abstract: Large language models (LLMs) excel at token-level generation but may learn undesirable abstract semantics and lack comprehensive perception. LLM-JEPA mitigates this by aligning different views of the same underlying knowledge via...

<https://arxiv.org/abs/2609.36952>

67 CONF

Controlled Decoding Attacks on Black-Box LLMs

ArXiv CS.AI 50%

arXiv:2609.36956v1 Announce Type: cross Abstract: Manipulating next-token probabilities during generation can bypass the safety alignment of large language models. Existing approaches, however, rely on access to model weights or numerical token probabilities and therefore do not...

<https://arxiv.org/abs/2609.36956>

68 CONF

NowcastDiT: Diffusion Transformers are Effective Precipitation Nowcasters

ArXiv CS.AI 50%

arXiv:2609.37038v1 Announce Type: cross Abstract: Precipitation nowcasting demands accurate short-term forecasts under strong spatiotemporal variability. Diffusion models are well suited to modeling complex precipitation distributions, yet existing approaches often introduce...

<https://arxiv.org/abs/2609.37038>

69 CONF

LazySloth: Bounded LLM-based Lazy Tree Search for Fast Long Video Comprehension

ArXiv CS.AI 50%

arXiv:2609.37426v1 Announce Type: cross Abstract: Modern vision-language models (VLMs) have shown promising results in long-video understanding due to the rich semantic information they can capture. However, most methods focus on coarse captioning of extracted image frames that...

<https://arxiv.org/abs/2609.37426>

70 CONF

A Comprehensive View of Fairness through Distributional Stability

ArXiv CS.AI 50%

arXiv:2609.37061v1 Announce Type: cross Abstract: We view fairness as a property of distributional stability. Rather than assessing a predictor under a fixed data distribution, we study how its predictions change under perturbations that modify the composition of protected...

<https://arxiv.org/abs/2609.37061>

71 CONF

Beyond Compression: Diagnosing How Post-Training Changes Mathematical Reasoning

ArXiv CS.AI 50%

arXiv:2609.37066v1 Announce Type: cross Abstract: Post-training is central to mathematical reasoning in modern large language models (LLMs), but endpoint pass@1 alone underidentifies what has changed. Gains may reflect newly reachable solutions, cheaper sampling of latent...

<https://arxiv.org/abs/2609.37066>

72 CONF

Designing a Boundary Negotiating Artifact for Collaborative Socio-Technical Sense-Making in AI Regulatory Sandboxes

ArXiv CS.AI 50%

arXiv:2609.37109v1 Announce Type: cross Abstract: The rapid, unpredictable advancements in AI system capabilities has seen regulators take adaptive and experimental approaches to policymaking. Established in other domains as instruments balancing regulation with innovation,...

<https://arxiv.org/abs/2609.37109>

73 CONF

Multichannel Audio Quality Assessment: Extending Pretrained Perceptual Models to Spatial Audio

ArXiv CS.AI 50%

arXiv:2609.37116v1 Announce Type: cross Abstract: Accurate perceptual quality assessment is essential for evaluating and optimizing spatial audio, where perceived quality depends on both signal fidelity and inter-channel spatial relationships. However, subjective evaluation is...

<https://arxiv.org/abs/2609.37116>

74 CONF

Trajectory Soup: Pushing the Compute-Scaling Frontier of LLM Mid-training via Diverse Trajectories

ArXiv CS.AI 50%

arXiv:2609.37169v1 Announce Type: cross Abstract: Mid-training equips pretrained large language models with specialized and reasoning capabilities, but the returns of this stage are bounded since additional serial compute yields little further downstream improvement and can even...

<https://arxiv.org/abs/2609.37169>

75 CONF

CRJudgeBench: Can AI Detect Plausible but Invalid Code Reviews?

ArXiv CS.AI 50%

arXiv:2609.37216v1 Announce Type: cross Abstract: Large language models can generate plausible code-review comments, but such comments may contain technically incorrect claims that mislead developers. We study technical trustworthiness judgment: determining whether a review...

<https://arxiv.org/abs/2609.37216>

76 CONF

Beyond Semantic Narrowing: Robust and Efficient LLM Watermarking with Hamming Neighborhoods

ArXiv CS.AI 50%

arXiv:2609.37218v1 Announce Type: cross Abstract: Semantic watermarking improves robustness against watermark removal attacks by embedding detectable signals into sentence-level representations. However, existing watermarking methods typically impose watermark-specific semantic...

<https://arxiv.org/abs/2609.37218>

77 CONF

Hidden Reasoning Must Leak, but Need Not Be Readable: Fundamental Opportunities and Limits for Chain-of-Thought Monitoring

ArXiv CS.AI 50%

arXiv:2609.37312v1 Announce Type: cross Abstract: Can reasoning models trick chain of thought (CoT) monitors and perform hidden computation without revealing it in their thinking traces? We show that the answer depends on the underlying task difficulty and the model size. Simple...

<https://arxiv.org/abs/2609.37312>

78 CONF

Do Agent Benchmarks Do What They Say? An Executable-Contract Audit of Tool-Using Agent Environments

ArXiv CS.AI 50%

arXiv:2609.37315v1 Announce Type: cross Abstract: Tool-using agents are entering settings where a wrong action carries real cost, and the benchmarks certifying them grade what each simulated tool call reports having done, assuming the tool did what its interface advertises. The...

<https://arxiv.org/abs/2609.37315>

79 CONF

TAEC: Trajectory-Aware Evidence Coordination for Multi-Step Visual RAG

ArXiv CS.AI 50%

arXiv:2609.37349v1 Announce Type: cross Abstract: Multi-step visual retrieval-augmented generation (RAG) answers complex questions by repeatedly retrieving visual evidence, updating an intermediate state, and deciding whether to continue searching or answer. Yet retrieving...

<https://arxiv.org/abs/2609.37349>

80 **CONF****Compiling Learning Problems into Adaptation Programs for Language Models****ArXiv CS.AI** **50%**

arXiv:2609.37371v1 Announce Type: cross Abstract: Model adaptation is typically governed by a fixed recipe, even though different update programs can produce substantially different behavioral outcomes. We introduce adaptation compilation, which reframes where, how, and to what...

<https://arxiv.org/abs/2609.37371>

81 **CONF****Complexity-Aware Evaluation of LLM Comprehension****ArXiv CS.AI** **50%**

arXiv:2609.37405v1 Announce Type: cross Abstract: Large language models (LLMs) are increasingly used for software engineering tasks that require understanding existing source code, including behavior prediction, function explanation, debugging, and code review. However,...

<https://arxiv.org/abs/2609.37405>

82 **CONF****Query Lower Bounds for Diffusion Sampling****ArXiv CS.AI** **50%**

arXiv:2604.10857v2 Announce Type: replace-cross Abstract: Diffusion models generate samples by iteratively querying learned score estimates. A rapidly growing literature focuses on accelerating sampling by minimizing the number of score evaluations, yet the information-theoretic...

<https://arxiv.org/abs/2604.10857>

83 **CONF****Risk-Controlled Selective LLM Answering by Pricing Label-Free Checks****ArXiv CS.AI** **50%**

arXiv:2609.37493v1 Announce Type: cross Abstract: Serving an answer from a large language model requires deciding when to abstain, yet a verifier's ranking accuracy alone does not determine the error rate among served answers. We introduce PriceCheck, which builds a compact...

<https://arxiv.org/abs/2609.37493>

84 **CONF****Evaluating Bounded Autonomy in Regulated Agentic AI: A Diagnostic Harness with Constitutional Rewards, Escalation Labels, and Runtime Governance****ArXiv CS.AI** **50%**

arXiv:2609.37501v1 Announce Type: cross Abstract: We propose RegLLM, a diagnostic harness for bounded autonomy in regulated agentic workflows. It instruments six trustworthiness signals: citation validity, source grounding, schema compliance, escalation correctness,...

<https://arxiv.org/abs/2609.37501>

85 **CONF****Risk-Aware Semantic Grounding for Trustworthy LLM-Based Robot Planning****ArXiv CS.AI** **50%**

arXiv:2609.37554v1 Announce Type: cross Abstract: Large language models (LLMs) are increasingly used as high-level planners in robot navigation, but their outputs may become unreliable when instructions are ambiguous, unsupported by the environment, or semantically inconsistent...

<https://arxiv.org/abs/2609.37554>

86 CONF

Concealing LLM-Based Multi-Agent Topology via Phantom Structure Injection

ArXiv CS.AI 50%

arXiv:2609.37567v1 Announce Type: cross Abstract: Driven by the rapid advancement of large language models (LLMs), LLM-based multi-agent systems (MAS) have emerged as a powerful paradigm for collaborative reasoning over complex tasks. A key design element of MAS is the...

<https://arxiv.org/abs/2609.37567>

87 CONF

FedSocket: Recipient-Executable Knowledge Exchange for Heterogeneous Multimodal Federated Learning

ArXiv CS.AI 50%

arXiv:2609.37582v1 Announce Type: cross Abstract: Federated knowledge must remain usable by recipients with different modalities, private architectures, and tasks. We present FedSocket, which makes recipient execution a design requirement of the exchanged model. A shared Q...

<https://arxiv.org/abs/2609.37582>

88 CONF

Independent Verification Paths Are Not Independent: A Case Study of Common-Mode Failure in a Satellite Catalogue Pipeline

ArXiv CS.AI 50%

arXiv:2609.37603v1 Announce Type: cross Abstract: A common safeguard for a data pipeline is redundant computation: derive each published number by two routes built on different technology and refuse to exit when they disagree. We report one such gate failing, in a...

<https://arxiv.org/abs/2609.37603>

89 CONF

Correct, Don't Delete: Mitigating Emergent Misalignment with Corrective Supervision

ArXiv CS.AI 50%

arXiv:2609.37624v1 Announce Type: cross Abstract: Fine-tuning a language model on a narrow set of harmful demonstrations, such as bad medical advice, can make it broadly misaligned on unrelated questions, a phenomenon known as emergent misalignment (EM). The usual defense is to...

<https://arxiv.org/abs/2609.37624>

90 CONF

Training-Free Global Geometric Association for 4D LiDAR Panoptic Segmentation

ArXiv CS.AI 50%

arXiv:2512.18991v3 Announce Type: replace-cross Abstract: Dominant paradigms for 4D LiDAR panoptic segmentation are usually required to train deep neural networks with large superimposed point clouds or design dedicated modules for instance association. However, these approaches...

<https://arxiv.org/abs/2512.18991>

91 CONF

Semantic Map Sharing and Capability-Aware Coverage Planning for AI-Native 6G Robotic Coordination

ArXiv CS.AI 50%

arXiv:2609.37666v1 Announce Type: cross Abstract: Search and Rescue (SAR) operations increasingly deploy heterogeneous teams of aerial and ground robots. However, conventional coverage methods typically do not translate perceived terrain into platform-specific reachability,...

<https://arxiv.org/abs/2609.37666>

92 CONF

Multi-Site Real-World Performance of Commercial AI for Pulmonary and Incidental Pulmonary Embolism Detection

ArXiv CS.AI 50%

arXiv:2609.37750v1 Announce Type: cross Abstract: Pulmonary embolism (PE) is a leading cause of cardiovascular mortality, yet the real-world performance of FDA-cleared AI detection models remains incompletely characterized. We retrospectively evaluated two FDA-cleared AI...

<https://arxiv.org/abs/2609.37750>

93 CONF

A Benchmark & Dataset for Detecting AI-Manipulated Visual Evidence in the Court System

ArXiv CS.AI 50%

arXiv:2609.37783v1 Announce Type: cross Abstract: Photographic evidence is becoming increasingly vulnerable to forms of alteration and fabrication that existing legal and technical workflows are not well equipped to evaluate. Surveillance frames, dashcam stills, and phone...

<https://arxiv.org/abs/2609.37783>

94 CONF

Pixel-Level Transformers in Remote Sensing: A Canopy Height Case Study

ArXiv CS.AI 50%

arXiv:2609.37809v1 Announce Type: cross Abstract: Predicting canopy height from medium-resolution satellite imagery is a common and scalable approach for assessing the condition of the world's forests, which play a crucial role in climate change mitigation. While...

<https://arxiv.org/abs/2609.37809>

95 CONF

Making Duplicate Reimbursement Unrepresentable: A Verified Ethereum E-Invoice System for Humans and AI Agents

ArXiv CS.AI 50%

arXiv:2609.37819v1 Announce Type: cross Abstract: Electronic invoices are replacing paper invoices worldwide, but today's centralized architectures leave three problems unsolved on the consumption side: an invoice can be submitted for reimbursement repeatedly, authenticity is...

<https://arxiv.org/abs/2609.37819>

96 CONF

Scaling Influence Functions in LLMs through Eigenbasis-Corrected One-Bit Gradient Projection

ArXiv CS.AI 50%

arXiv:2609.37842v1 Announce Type: cross Abstract: Influence functions estimate how individual training examples affect the behavior of large language models (LLMs). Analyzing how training data influence different behaviors of an LLM involves repeated influence computation....

<https://arxiv.org/abs/2609.37842>

97 CONF

ExceptionDrive: A Planning-Oriented Counterfactual Corner-Case Benchmark for Autonomous Driving

ArXiv CS.AI 50%

arXiv:2609.37871v1 Announce Type: cross Abstract: Average performance on routine driving benchmarks does not establish planner reliability under rare, safety-critical hazards. We proposed ExceptionDrive, a counterfactual planning benchmark that uses VLM-assisted screening,...

<https://arxiv.org/abs/2609.37871>

98 CONF

The Unequal Influence of Bad Advice: Using Training Data Attribution to Modulate Emergent Misalignment

ArXiv CS.AI 50%

arXiv:2609.37914v1 Announce Type: cross Abstract: Fine-tuning large language models on narrow, misaligned tasks can undo their post-training alignment and induce novel misaligned behaviors -- a phenomenon known as \emph{emergent misalignment} (EM). EM has been linked to...

<https://arxiv.org/abs/2609.37914>

99 CONF

Does Local Video Understanding Transfer Across Encounters? The EgoGears Benchmark

ArXiv CS.AI 50%

arXiv:2609.37938v1 Announce Type: cross Abstract: Embodied systems must make knowledge acquired during one encounter usable in another despite changes in viewpoint, motion, and illumination. Yet aggregate cross-video accuracy conflates failures of local perception with failures...

<https://arxiv.org/abs/2609.37938>

00 CONF

Dagger: Decoupling-based Model Stealing Attack against Graph Neural Networks

ArXiv CS.AI 50%

arXiv:2609.37972v1 Announce Type: cross Abstract: As Graph Neural Networks (GNNs) are widely deployed as Machine Learning-as-a-Service (MLaaS) APIs, model stealing attacks have emerged as a critical security threat. By querying a victim model's black-box API, an adversary can...

<https://arxiv.org/abs/2609.37972>

01 CONF

From Unity Simulation to Diffusion-Based Augmentation: Quantifying Dataset Balance for Robust Object Detection

ArXiv CS.AI 50%

arXiv:2609.38010v1 Announce Type: cross Abstract: Modern computer vision models achieve high accuracy when trained on large-scale annotated datasets. In critical domains such as construction safety monitoring, data collection is costly, hazardous, and ethically constrained. This...

<https://arxiv.org/abs/2609.38010>

02 CONF

Auditable Long-Term Memory: A Deterministic Retrieval Chain Measured at 479/475 of 500 on LongMemEval-S

ArXiv CS.AI 50%

arXiv:2609.38021v1 Announce Type: cross Abstract: We evaluate an auditable long-term memory system on LongMemEval-S. Its retrieval chain uses hybrid candidate retrieval, cross-encoder reranking, coverage-first packet compilation, and deterministic reasoning scaffolds; an LLM is...

<https://arxiv.org/abs/2609.38021>

03 CONF

Gender bias across LLMs is common and highly heterogenous

ArXiv CS.AI 50%

arXiv:2609.38036v1 Announce Type: cross Abstract: Understanding gender biases in large language models (LLMs) is increasingly important as these systems become embedded in decision-support tools with real consequences. Prior research has focused only on a small set of models,...

<https://arxiv.org/abs/2609.38036>

04 CONF

Correct Answers, Invalid Traces: What Verifiable Grade-School Math Reveals About Chain-of-Thought Traces

ArXiv CS.AI 50%

arXiv:2609.38107v1 Announce Type: cross Abstract: Chain-of-thought traces are widely read as records of how models reach their answers, informing debugging, agent auditing, and claims about reasoning. Testing this interpretation is difficult because natural-language thinking...

<https://arxiv.org/abs/2609.38107>

05 CONF

Breaking the Uniformity Trap: Scaling Video Diffusion Model via SplitMoE

ArXiv CS.AI 50%

arXiv:2609.38140v1 Announce Type: cross Abstract: Mixture-of-Experts (MoE), popularized by large language models, is a promising paradigm for scaling visual generative models. However, conventional token-wise MoE routes tokens independently within a homogeneous expert pool and...

<https://arxiv.org/abs/2609.38140>

06 CONF

Just Initialize: A Training-Free Initialization Component for Large-Scale Routing Optimization

ArXiv CS.AI 50%

arXiv:2609.35443v2 Announce Type: replace Abstract: Large-scale routing problems are difficult to solve efficiently as their search spaces grow rapidly with problem size. Existing approaches primarily improve the optimization procedure itself, often at increasing computational...

<https://arxiv.org/abs/2609.35443>

07 CONF

FloCA: Towards Faithful and Logically Consistent Flowchart Reasoning

ArXiv CS.AI 50%

arXiv:2602.14035v2 Announce Type: replace Abstract: Flowchart-oriented dialogue (FOD) systems aim to guide users through multi-turn decision-making or operational procedures by following a domain-specific flowchart to achieve a task goal. In this work, we formalize flowchart...

<https://arxiv.org/abs/2602.14035>

08 CONF

One Readout, Many Repairs: Diffusion-Guided Hierarchical Search for Tool-Agent Repair

ArXiv CS.AI 50%

arXiv:2609.34879v2 Announce Type: replace Abstract: Tool agents use large language models to act through external tools, yet successfully executed calls can still leave user requests unfulfilled. Tool-agent repair seeks alternative call sequences that execute successfully and...

<https://arxiv.org/abs/2609.34879>

09 CONF

How Should Reasoning Be Organized in a Transformer's Latent Space?

ArXiv CS.AI 50%

arXiv:2609.13747v2 Announce Type: replace Abstract: Continuous reasoning has emerged as a promising way to improve reasoning in large language models (LLMs). Yet we still lack a clear principle for deciding what a latent state should preserve. Reasoning by superposition shows...

<https://arxiv.org/abs/2609.13747>

10 CONF

Do Proactive Agents Need an LLM to Decide When to Act?

ArXiv CS.AI 50%

arXiv:2605.30152v2 Announce Type: replace-cross Abstract: Proactive assistants continuously decide when to intervene and what context should support the intervention. Large language model (LLM) pipelines repeatedly interpret activity histories to make these decisions, paying an...

<https://arxiv.org/abs/2605.30152>

11 CONF

The Commit-Abstain Circuit: Why Language Models Hallucinate Instead of Abstaining

ArXiv CS.AI 50%

arXiv:2609.32964v2 Announce Type: replace Abstract: Language models (LMs) often hallucinate by committing to confident answers rather than abstaining, even when they do not have enough information to answer reliably. A large body of existing work mitigates hallucination through...

<https://arxiv.org/abs/2609.32964>

12 CONF

Compositional Safety Failures in Harness Evolution: Identification and Runtime Monitoring

ArXiv CS.AI 50%

arXiv:2609.33123v2 Announce Type: replace Abstract: Self-evolving agent harnesses continually update persistent components such as memory, prompts, skills, and tools. We call this process harness evolution. However, such evolution could introduce unexpected safety risks....

<https://arxiv.org/abs/2609.33123>

13 CONF

GUITAR: Structured Failure Diagnosis of GUI Agents via State Transitions

ArXiv CS.AI 50%

arXiv:2609.34113v2 Announce Type: replace Abstract: Understanding where and why Graphical User Interface (GUI) agents fail is essential for building more reliable systems, yet current evaluation relies on step accuracy, a metric that treats each screen independently and...

<https://arxiv.org/abs/2609.34113>

14 CONF

OmniTide: Co-Designing Algorithms and Systems for Efficient On-Device Omni-LLM Streaming

ArXiv CS.AI 50%

arXiv:2609.34653v2 Announce Type: replace Abstract: On-device streaming omni-modal inference safeguards user privacy and eliminates prohibitive per-token API costs, but faces a critical bottleneck: the continuous influx of multimodal data rapidly exhausts constrained memory and...

<https://arxiv.org/abs/2609.34653>

15 CONF

Privacy-Preserving Full-Body Meshing from mmWave Radar via Mesh Foundation Model Supervision

ArXiv CS.AI 50%

arXiv:2609.34768v2 Announce Type: replace Abstract: Millimeter-wave (mmWave) radar enables privacy-preserving human perception, but the extreme sparsity of point clouds from commercial single-chip sensors (mean ~6.5 points/frame; ~28% empty frames) has confined prior art to...

<https://arxiv.org/abs/2609.34768>

16 CONF

STRIDE: Automated Evaluation of Text-to-Trajectory Alignment across Diverse Contexts

ArXiv CS.AI 50%

arXiv:2609.34799v2 Announce Type: replace Abstract: Language-conditioned trajectory generation is here, but its evaluation has not kept pace. Existing pedestrian trajectory metrics compare trajectories with real-world human data. This does not scale to text-to-trajectory...

<https://arxiv.org/abs/2609.34799>

17 CONF

Building Transformation Layers for Riemannian Neural Networks

ArXiv CS.AI 50%

arXiv:2609.35436v2 Announce Type: replace Abstract: Recently, deep neural networks on manifold-valued representations have garnered significant attention across various machine learning applications. One recent focus is the generalization of Euclidean fully connected (FC) and...

<https://arxiv.org/abs/2609.35436>

18 CONF

On the Approximation and Convergence of Distributional Policy Gradient Algorithms for Risk-Sensitive Reinforcement Learning

ArXiv CS.AI 50%

arXiv:2405.14749v3 Announce Type: replace-cross Abstract: Risk-sensitive reinforcement learning (RL) is crucial for maintaining reliable performance in high-stakes applications. While traditional RL methods aim to learn a point estimate of the random cumulative cost...

<https://arxiv.org/abs/2405.14749>

19 CONF

Causal pieces: analysing and improving spiking neural networks piece by piece

ArXiv CS.AI 50%

arXiv:2504.14015v2 Announce Type: replace-cross Abstract: We introduce "causal pieces", a novel concept for analysing spiking neural networks (SNNs), inspired by "linear pieces" used to study expressivity and trainability in artificial neural networks (ANNs). Causal pieces...

<https://arxiv.org/abs/2504.14015>

20 CONF

Uncertainty-aware Causal Decision Making via Effect Bound Decomposition

ArXiv CS.AI 50%

arXiv:2601.22736v3 Announce Type: replace-cross Abstract: Causal inference from observational data can provide strong evidence for finding the best action in a decision-making scenario without having to perform expensive randomized trials. The causal effect of an action is often...

<https://arxiv.org/abs/2601.22736>

21 CONF

Agentic AI for Scalable and Robust Optical Systems Control

ArXiv CS.AI 50%

arXiv:2602.20144v2 Announce Type: replace-cross Abstract: We present AgentOptics, an agentic AI framework for high-fidelity, autonomous optical system control built on the Model Context Protocol (MCP). AgentOptics interprets natural language tasks and executes protocol-compliant...

<https://arxiv.org/abs/2602.20144>

22 CONF

Altered Thoughts, Altered Actions: Reasoning Chain as Control Surface for a Vision-Language-Action Policy

ArXiv CS.AI 50%

arXiv:2603.12717v2 Announce Type: replace-cross Abstract: Vision-language-action policies map camera images and natural-language instructions to a robot's motor actions. Some of these policies are designed to reason in text before acting, generating a reasoning chain and...

<https://arxiv.org/abs/2603.12717>

23 CONF

Pure and physics-guided deep learning approaches for spatio-temporal groundwater level prediction

ArXiv CS.AI 50%

arXiv:2603.25779v2 Announce Type: replace-cross Abstract: Groundwater represents a key element of the water cycle, yet it exhibits complex and context-dependent relationships that make its modeling challenging. Theory-based models have been the cornerstone of scientific...

<https://arxiv.org/abs/2603.25779>

24 CONF

PAC-CF: Calibrating Irreversible Frontier Pruning in LLM-Guided Search

ArXiv CS.AI 50%

arXiv:2604.14345v5 Announce Type: replace-cross Abstract: LLM-guided search explores multiple candidate trajectories, but at substantial test-time cost. Pruning low-scoring frontier candidates can control this cost, yet it also turns potentially biased evaluator scores into...

<https://arxiv.org/abs/2604.14345>

25 CONF

Single-turn emergency psychiatric triage across 15 frontier AI chatbots

ArXiv CS.AI 50%

arXiv:2604.25415v2 Announce Type: replace-cross Abstract: People increasingly turn to general-purpose AI chatbots for advice about emotional and mental health problems, but the ability of these systems to recognize and appropriately triage psychiatric emergencies remains...

<https://arxiv.org/abs/2604.25415>

26 CONF

Efficient Pre-Training of LLMs through Truncated SVD Representations

ArXiv CS.AI 50%

arXiv:2605.28573v2 Announce Type: replace-cross Abstract: LLM pretraining is extremely costly; therefore, parameter-efficient LLM architectures have recently emerged as a compelling research direction. One such promising approach is to represent the parameters as orthonormal...

<https://arxiv.org/abs/2605.28573>

27 CONF

Is Agent Code Less Maintainable Than Human Code?

ArXiv CS.AI 50%

arXiv:2606.21804v2 Announce Type: replace-cross Abstract: Maintainability is a core dimension of software engineering, shaping how code is written, reviewed, and developed over time. While coding agents have demonstrated strong performance on single-issue tasks, it remains...

<https://arxiv.org/abs/2606.21804>

28 CONF

Low-power analogue neural networks with trainable nonlinear connections for continuous control

ArXiv CS.AI 50%

arXiv:2606.23742v2 Announce Type: replace-cross Abstract: Physical neural networks promise low-power machine learning by computing directly with analogue device physics, but most architectures force nonlinear device responses to act as scalar weights. Inspired by...

<https://arxiv.org/abs/2606.23742>

29 CONF

Does Anthropomorphic Language Impact Public Perceptions of AI?

ArXiv CS.AI 50%

arXiv:2606.29121v2 Announce Type: replace-cross Abstract: Public discourse about artificial intelligence (AI) often uses anthropomorphic language: language that attributes human capabilities and characteristics to AI systems. This practice has been criticized for setting...

<https://arxiv.org/abs/2606.29121>

30 CONF

Homo-RAG: Homology-Guided Retrieval-Augmented Generation for Cross-Species Gene Function Prediction

ArXiv CS.AI 50%

arXiv:2608.25466v2 Announce Type: replace-cross Abstract: The functional annotation of genes in non-model organisms remains a significant challenge in computational biology, with 20-70% of sequenced genes lacking characterized functions. Traditional homology-based methods are...

<https://arxiv.org/abs/2608.25466>

31 CONF

Kalman Delta Networks: Uncertainty-aware Associative Memory

ArXiv CS.AI 50%

arXiv:2609.07816v2 Announce Type: replace-cross Abstract: Linear attention enables efficient long-context inference by compressing token history into a fixed-size recurrent memory. This compression makes each update a trade-off between incorporating new information and...

<https://arxiv.org/abs/2609.07816>

32 CONF

ASCEND: Personal AI Agents for Autonomous Scientific Computing Across HPC Clusters and GPU Workstations

ArXiv CS.AI 50%

arXiv:2609.32868v2 Announce Type: replace-cross Abstract: Traditional scientific computing requires researchers to translate computational intent into environment configuration, resource requests, and executable jobs, then diagnose failures from scheduler state and application...

<https://arxiv.org/abs/2609.32868>

33 CONF

Multimodal LLMs Outperform Pathology Foundation Models in Cross-Domain Histological Similarity

ArXiv CS.AI 50%

arXiv:2609.32876v2 Announce Type: replace-cross Abstract: State-of-the-art pathology foundation models, trained on millions of histology tiles, can fail to preserve tissue similarity when comparisons cross slide or institution boundaries. We show that general-purpose multimodal...

<https://arxiv.org/abs/2609.32876>

34 CONF

Don't Throw Away the Tail: Action Upcycling for Policy Acceleration

ArXiv CS.AI 50%

arXiv:2609.34911v2 Announce Type: replace-cross Abstract: Modern robot policies predict a chunk of future actions from a single observation, execute only a prefix, and discard the rest before replanning. Choosing the length of this prefix, the execution horizon, poses a...
<https://arxiv.org/abs/2609.34911>

35 CONF

F4R: Failure-Driven Recognition, Reconstruction, Refinement, and Redeployment for Continual Robot Self-Improvement

ArXiv CS.AI 50%

arXiv:2609.35575v2 Announce Type: replace-cross Abstract: The real-world performance of current vision-language-action models is fundamentally constrained by the limited coverage of expert demonstrations and their insufficient understanding of physical interactions. A common...
<https://arxiv.org/abs/2609.35575>